8.5 Activities: estimating and the calculator

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1. Supermarket estimator

You need: a real shopping trip, or a receipt and a supermarket website.

As items go into the basket, he keeps a rough running total in his head, rounding each price to the nearest euro (€2,95 → €3, €1,19 → €1). At the till, compare his estimate with the real total. Over a few weeks, see how close he gets.

Talk about it: "Did your estimate come out above or below the real total? Why might that be?" (If most prices end in ,95 or ,99, rounding up makes the estimate a little high, which is a safe way to make sure you have enough money.)

2. Closest wins

You need: a calculator, paper, two or more players.

Someone says a sum such as 38 × 21. Everyone writes an estimate (no paper working, just friendly numbers). Then the calculator reveals the answer. Closest estimate scores a point. Note: the point is for a sensible estimate, not the exact answer; the parent should play too and sometimes lose.

Talk about it: "Which friendly numbers did you use? Would 40 × 20 or 38 × 20 be easier?"

3. Catch the calculator

You need: a calculator, a list of sums.

The parent "plays the calculator" and gives answers, some right and some silly (456 + 389 = 7 145; 2 500 : 50 = 5). He must catch the wrong ones with an estimate, without calculating exactly. Then swap roles: he invents wrong answers to catch you out.

Talk about it: "How did you know that one was wrong without working it out?"

4. Head, paper or calculator?

You need: 12 cards with sums on them (for example: 100 − 1; 25 × 4; 3 648 : 12; 487 + 235; 17,8% of €2 345; 999 + 999).

He sorts the cards into three piles: head, paper, calculator. There are no fixed right answers: one person's "head" is another's "paper". What matters is the reason. Then do a few from each pile.

Talk about it: "Why did you put 999 + 999 in the head pile?" (1 000 + 1 000 − 2.)

5. Calculator tour

You need: a basic calculator (a phone calculator works).

Explore together:

- Type 1 : 3. What do you see? Why does it go on and on?
- Type 7 : 2. The screen says 3.5. How do we write that at school? (3,5)
- Type 3 + 4 × 2 on a basic calculator. Then do 4 × 2 + 3. Same answer? Why not?
- Type 0.1 + 0.2. Then 10 × 0,1. Talk about the point key.
- Find the "C" or "AC" key. When do you use it?

Talk about it: "What's one way a calculator could trick you if you're not careful?"

6. How many…? estimation jar

You need: a jar with pasta, buttons, or sweets.

He estimates how many are inside, explaining a method (for example: count one layer and multiply by the number of layers). Then count. Change the contents each week.

Talk about it: “What would make a better estimate next time?”

Conversation prompts for any day

- “About how much will the shopping cost? Do we have enough with €20?”
- “The navigation says 47 minutes. Is that about three quarters of an hour or nearly an hour?”
- “The calculator says 12.5. What does that mean if we’re talking about money? About people?”
- “Would you do this one in your head, on paper, or on the calculator?”